

# Automatic Identification of Medicinal Plants with Transfer Learning

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**Abstract**—Medicinal plants are widely recognized for their medicinal properties, so accurate identification is very important for scientists, pharmacists and people. This work presents a transfer learning-based solution to identifying medicinal plants using MobileNetV2 architecture. Transfer learning can effectively classify plant species using pre-trained models through optimization of plant data. The system is designed to take user input images, process images from the MobileNetV2 model, and predict vegetation with high accuracy. After categorization, the system integrates the Google Custom Search API to provide useful information to users, including the top three websites describing the use of medicinal herbs. This article describes the architecture, processes, and procedures involved in developing Web applications. Our method simplifies the plant identification process using adaptive learning, thereby reducing training time and computational resources while increasing classification accuracy. The solution is designed to support veterinarians, teachers, and assistants in quickly identifying medicinal plants and understanding their uses, thus improving the overall understanding of phytotherapy.

**Keywords:** Transfer Learning, Medicinal Plant Identification, MobileNetV2, Image Classification, Google Custom Search API, Deep Learning, Web Application, Plant Species Classification

## I. INTRODUCTION

Medicinal plants have been an imperative portion of human wellbeing for thousands of a long time and have given treatment for numerous infections. According to traditional practices, their priority is that approximately 80% of the country's population is dependent on medicinal plants obtained from different plants such as leaves, roots and leaves. However, with increasing urbanization and loss of biodiversity, identification of these plants has become a challenge for researchers and practitioners in the herbal field.

Normally, the identification of medicinal plants requires knowledge and skills that are often rare. This deficiency is exacerbated by the complexity of plant morphology and the diversity of the environment in which plants grow, making manual work difficult and error-prone. In this context, the integration of computer vision and machine learning innovation has developed as a promising strategy for programmed plant identification. Recent developments in

deep learning, especially the use of neural networks (CNN), have shown great potential in accurately classifying plants based on visual features.

In this paper, we propose a transfer learning approach using MobileNetV2 architecture to identify medicinal plants. Our method can classify plant species with limited information using pre-trained models, thus optimizing input parameters while maintaining accuracy. The system is designed to integrate with Google's custom search API, allowing users to download images and provide useful information to complete real-time identification of plants. The API provides users with access to the first three websites that explain the uses and benefits of the identified drugs.

Our method not only increases the accuracy of plant identification, but also simplifies the process, making it accessible to physicians, teachers, and enthusiasts. This research focuses on the growing need for plant identification tools that can facilitate herbal medicine and contribute to a deeper understanding of medicinal plants. The following sections discuss the methodology, results, and implications of our findings, and highlight the role of educational change in improving classification.

To illustrate the diverse range of medicinal plants identified by our transfer learning-based approach, several examples of images from our dataset are presented in Figure 1.



Doddapatre      Hibiscus      Mint

Fig. 1: Sample Images

## II. LITERATURE SURVEY

Kavitha and her team investigated different CNN models to identify medicinal plants using the Indian plant dataset. They tested models like MobileNet, ResNet50, DenseNet121, Xception, and Inception v3; Inception v3 stood out and achieved 95.16% accuracy. They used data augmentation to expand the dataset to improve performance and stochastic gradient descent (SGD) to refine the model.

The data was split into 70% for training and 30% for testing. However, the limitation of their study is that it focused only on the features of leaves and ignored other important aspects of the plants like flowers and roots [1].

(Ghosh et al) developed a hybrid model combining PCA and VGG16 to identify medicinal plants. They used data for 30 plant species from the Indian subcontinent, including 60-100 images of each plant. To make the model more effective, they converted the image to grayscale and used PCA to reduce the data provided in VGG16 for classification. After training the model for 10, 25 and 50 times, they achieved 95.25% testing rate in 50 times. This model focuses on the important features of leaves such as area, roundness and texture. While this model is better than other models such as ResNet50 and Inception V3, it requires a lot of memory, which can be an issue with insufficient equipment. It also only focuses on leaves, ignoring other important parts of the plant such as flowers or bark [2].

Khalid and Romle conducted research using the MYLPherbs-1 dataset, which contains 2,093 images of two types of plants: *Centella Asiatica* and *Gynura Pseudochina*. They used the ResNet-50 architecture, previously trained on the ImageNet dataset, to classify these plants. Their approach involves the use of adaptive learning and optimization to improve segmentation. The model follows the characteristic extractor and is fine-tuned specifically for Level 4, Block 1. However, the study was limited to two plant species, which affects the generalizability of the model to other species. The optimization process also requires a lot of computation, and they found that changing the size had little effect on performance, indicating that more research is needed to improve it [3].

(Duong-Trung et al) developed a model to classify medicinal plants using MobileNet and transfer learning, focusing on accurate measurement with limited computational resources. They used data from 2296 plant specimens from 10 species in southern Vietnam; they classified the data for training, validation, and testing. By improving MobileNet using a deep segmentation algorithm, they achieved 98.7% accuracy in the benchmark while maintaining the deep model. However, the model showed signs of being outdated, perhaps due to the small amount of data, and the authors believe that the model can be improved with more data. They also talked about their future plans to transform the model into a mobile application [4].

(Kavitha et al.) trained a deep learning model using MobileNet to identify six medicinal plants. They used a Kaggle dataset containing 500 images of each type. Resize and enhance the images with techniques such as rotation and flip to improve the model performance. The model performed well, achieving 98.33% accuracy, 99.32% precision, and 97.36% recall on the test data. MobileNet's architecture uses a deep distributed network to help reduce computation, allowing it to operate efficiently in resource-hungry environments. The model has also been incorporated into a mobile application for real-time plant identification on Google Cloud. However, the study was limited to six plant species, which may impact its widespread use [5].

(Mulugeta et al.) conducted a systematic review to investigate the use of deep learning technology to classify medicinal plants. They analyzed 31 studies selected from a total of 1,644 studies published between January 2018 and December 2022. 64.5% of the transformation studies used convolutional neural networks (CNN) for distribution. Most of the studies focused on identifying the leaves of the plant and relied on specific information (67.7%). Further developing the technology. However, the main problem identified is the lack of universally valid population data on medicinal plants. The researchers also noted that it is important to make these patterns easier to interpret and more reliable when used for plant identification [6].

The author used a database of 3000 images covering 30 different medicinal plant species and used the MobileNetV2 model to identify them. They preprocessed the image by resizing it to 224x224 pixels and using techniques such as rotation and Gaussian blur to build a better model. By freezing some layers of the MobileNetV2 pre-training and adding new training layers, the model was able to reach 98.05% performance after 50 training sessions. While the template works well, its limited data and focus on page images mean that it will not work well in many real-world situations [7].

(Sachar et al.) investigated the classification of plants using three datasets (Swedish (15 species), Flavia (33 species) and Malaya Kew (44 species)) using adaptive learning. They tested four CNN models (VGG-16, Xception, MobileNetV2 and DenseNet121) for extraction and used random forest for classification. Among them, DenseNet121 performed the best and achieved accuracy (100%) on Swedish dataset, 99% on Flavia and 92.4% on Malaya Kew. However, this study only focused on leaf images and MalayaKew did not perform image enhancement, which may limit the results. Additionally, DenseNet121 requires high computing power, making it less efficient to use in resource-constrained environments [8].

### III. PROPOSED SYSTEM

Our system uses deep learning models and image processing tools to help users instantly identify medicinal plants quickly and accurately. This makes it easy for everyone from doctors to plant enthusiasts to access important information about this plant. At the core of our system is the MobileNetV2 architecture, designed to efficiently distribute plant images and enable rapid identification of different plants. Users can easily upload photos of medicinal plants through a user-friendly and intuitive interface. It is planned that good ideas will emerge for the MobileNetV2 model specially trained on medicinal plant data when images are submitted. This function is a step forward for uploaded images. This includes resizing them, normalizing pixel values, and using object enhancement. Once the image is prepared, the MobileNetV2 model will analyze the image, identify plants, and quickly send the results to the user. Information about the plants they detect. The API has three main websites that explain the uses and benefits of each plant species and help users understand how to use them in real life. Time and skills required for

plant identification. Our system combines deep learning and network technology to enable users to quickly and accurately learn about medicinal plants, providing a deeper understanding of treatment facilities.

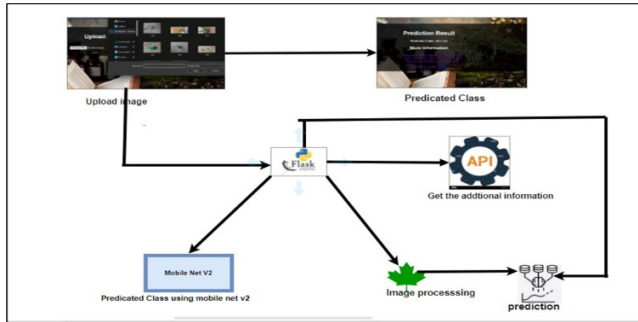


Fig. 2: System Architecture Diagram

### A. Data Collection

The data used in this study is taken from the Indian Medicinal Images database available on Kaggle, which contains a total of 4,765 images of medicinal plants with 40 different categories. Each category corresponds to a particular species and the images are carefully selected to ensure diversity in the collection. Use TensorFlow and Keras to load the data for further processing.

### B. Data Visualization and Distribution Analysis

To understand the distribution of classes within the dataset, visualizations were created using Matplotlib and Seaborn. A count plot was generated to showcase the number of images per class, which helps identify any imbalances that could affect model training. Figure 3 and Figure 4 shows

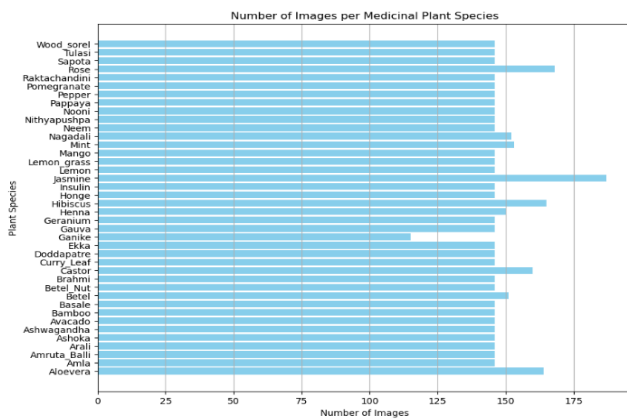


Fig. 3: Output of Count Plot Method

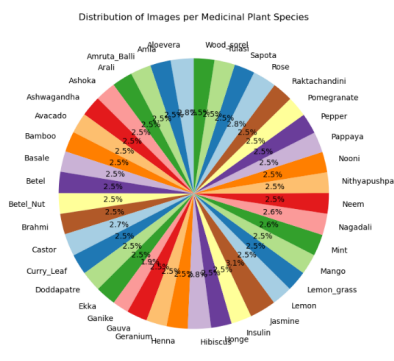


Fig. 4: Distribution Analysis

### C. Data Preprocessing

Images in the data often contain noise that may affect the quality of the training model. First, the following steps are performed:

1. *Image Resizing*: All images are resized to a single dimension (e.g. 224x224 pixels) to ensure a consistent input size for the MobileNetV2 model.
2. *Data Augmentation*: Use techniques such as rotation, scaling, and horizontal flip to increase the diversity of the datasets and improve model quality.
3. *Normalization*: Normalize the image pixel values to the range [0, 1] to facilitate better model learning.

### D. Feature Extraction Using Transfer Learning

Transfer learning was employed using the MobileNetV2 architecture, which is pre-trained on the ImageNet dataset. The last layers of the model were modified to adapt to the specific plant classification task.

The feature extraction process involves:

1. Loading the pre-trained MobileNetV2 model without the top layers.
2. Adding custom dense layers to classify the 40 different plant species.
3. Freezing the initial layers to retain learned features while training the new layers.

### E. Split the Dataset

The dataset is divided into training, validation, and test using an 80-20 ratio for training and validation. The testing procedure separates the training samples and the final evaluation after optimization. This ensures that the model is trained on a large portion of the data, while also keeping some for tuning the hyperparameters and evaluating the generalization ability of the model.

The splitting can be expressed as:

$$\text{Dataset} = \{X_{\text{train}}, Y_{\text{train}}, X_{\text{val}}, Y_{\text{val}}\}$$

Where:

- (a)  $X_{\text{train}}$  is the training set (80% of the data).
- (b)  $X_{\text{val}}$  is the validation set (20% of the data).
- (c)  $Y_{\text{train}}, Y_{\text{val}}$  represent the corresponding label sets for training, validation respectively.

### F. Model Training

Model training using working modifications of Mobilenetv2 architecture and previously trained on Imagenet dataset. The training process consists of the following steps:

1. *Data refinement and progression*: Use ImageDataGenerator to refine data by rescaling, cropping, zooming, and horizontal flipping, keeping 20% of the data for validation.
2. *Model design*: The MobileNetV2 model is loaded without an upper layer based on the global average layer, the 1024-neuron network layer, and the SoftMax layer corresponding to the facility composition.
3. *Model Collection*: The model was built using the Adam optimizer and categorical cross-entropy loss.
4. *Training*: First, the model is trained 20 times using the training generator and validated on the stored data.



- 5. *Fine-tuning*: The last 20 layers of the base model are not frozen for fine-tuning, and the model is iterated 10 more times with a lower learning rate of 0.00001.
- 6. *Model saved*: The training model is saved as `treatment_plant_classifier.h5` for future reference.

G. Model Evaluation

After training, assess the interpretation of the model in testing by calculating the correct score (i.e. the ratio of correct predictions to all predictions). This score helps to evaluate the effectiveness of the model in distinguish pharmaceutical plants based on input images. Our model achieved 93% accuracy, demonstrating good knowledge in classifying images of 40 different plant species.

IV. RESULT AND DISCUSSION

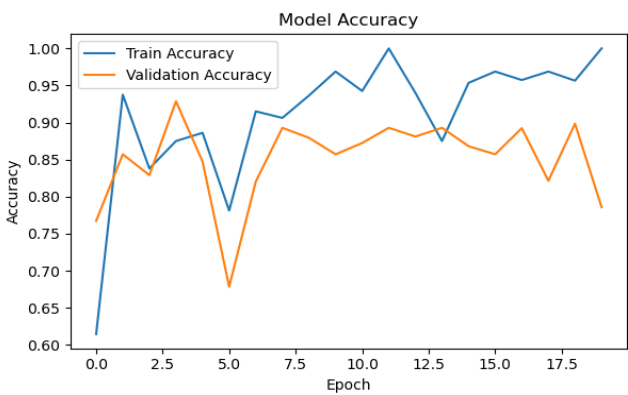


Fig. 5: Model Accuracy

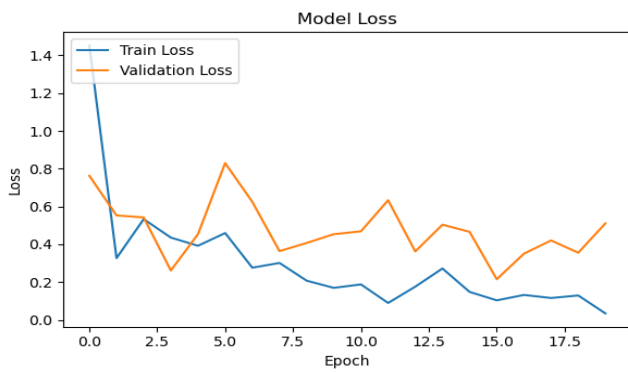


Fig. 6: Model Loss

The graph in Figure 5 above illustrates how the model is performing in terms of accuracy and Figure 6 depicts the loss with epochs related to medical plant classification in the form of a graph.

Table 1: Comparison of Medical Plant Classification Method

| No | Paper Name                                                       | Model Used                                | Testing Accuracy                                    |
|----|------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------|
| 1  | Identification of Medicinal Plant using Hybrid Transfer Learning | Hybrid PCA-VGG16<br>VGG16                 | 95.25%<br>76%                                       |
| 2  | Automatic Plant Identification Using Transfer Learning           | DenseNet121<br>DenseNet121<br>DenseNet121 | 100% (Swedish)<br>99% (Flavia)<br>92.4% (MalayaKew) |

|    |                                                                                          |                               |                 |
|----|------------------------------------------------------------------------------------------|-------------------------------|-----------------|
| 3  | A Combination of Transfer Learning and Deep Learning for Medicinal Plant Classification  | MobileNet                     | 98.7%           |
| 4  | Identification of Medicinal Plants in Ardabil Using Deep Learning                        | MobileNetV2                   | 98.05%          |
| 5  | Medicinal Plant Species Detection using Deep Learning                                    | Inception v3                  | 95%             |
| 6  | Herbal Plant Image Classification using Transfer Learning                                | ResNet-50 (Feature Extractor) | 88.45% - 93.18% |
| 7  | Medicinal Plant Identification in Real-Time Using Deep Learning Model                    | MobileNet                     | 97%             |
| 8  | Real-Time Identification of Medicinal Plants using Machine Learning                      | Logistic Regression           | 93.3%           |
| 9  | Identification of Different Plants through Image Processing                              | SVM                           | 95.17%          |
| 10 | An Application of Transfer Learning Techniques in Identifying Herbal Plants in Sri Lanka | ResNet                        | 95.5%           |
| 11 | Automatic Identification of Medicinal Plants with Transfer Learning                      | MobileNet V2                  | 93.75%          |

Above Table 1 shows comparative analysis between different models and algorithms. And the Figure 6 illustrate Comparison of plant identification models.

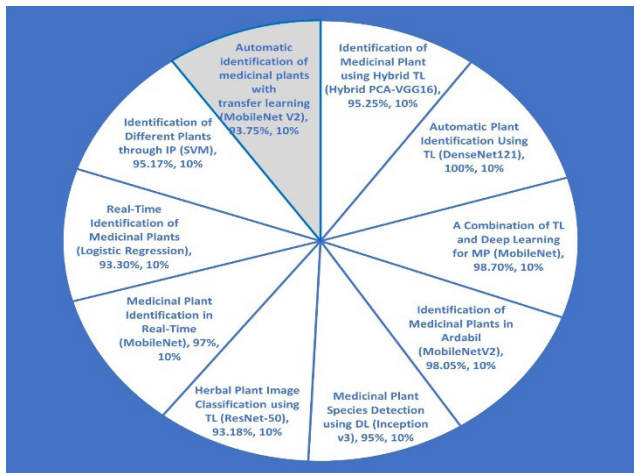


Fig. 6: Comparison of Plant Identification Models

V. CONCLUSION AND FUTURE SCOPE

Our medicinal plant identification system uses the MobileNetV2 model to provide robust classification of plant images into 40 different species. The model uses adaptive learning technology to improve plant information to achieve 93% accuracy. The integration of Google’s Custom Search

API improves the user experience by providing valuable information about identified plants, thus encouraging better understanding and use of medicinal herbs. The system not only simplifies the identification process, but also makes it accessible to many users, including doctors, teachers, and herbalists.

To improve the model's capabilities, future work could focus on expanding the data to include different plant species and images taken under different conditions, which would increase the strength of the structure. In addition, the development of mobile applications would facilitate instantaneous analysis and allow users to capture and analyze images anytime and anywhere. Using a feedback system could improve the model over time by adding user feedback or additional information. In addition, the integration of cloud-based AI services could increase the speed of operation and storage capacity, while optimizing low-cost space would also increase efficiency.

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